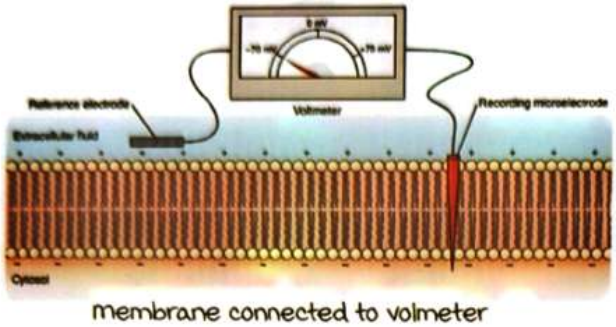
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# MEMBRANE POTENTIALS

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Membrane connected to volmeter

Types :

1. Resting membrane Potential (RMP).
2. Equilibrium Potential (EP).

## Resting Membrane Potential

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AKA Diffusion potential (D/t diffusion of $K^+$).

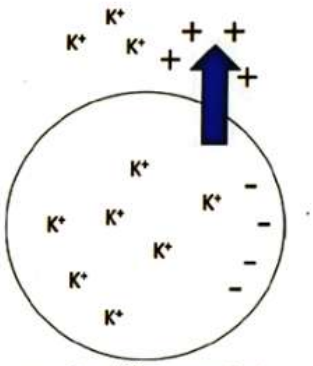
**FACTORS DETERMINING NEGATIVE RMP :**

1. Charge : Positively charged $K^+$ ion move out of cell.
2. Concentration gradient of $K^+$ :
   - Intracellular concentration : High.
   - Extracellular concentration : Low.
3. Permeability of cell to $K^+$ at rest :

Passive diffusion of $K^+$

↓

↑ Intracellular negative charge




Simple diffusion of $K^+$

**TYPES :**

**Stable :**

| Cell | Resting membrane potential |
|---|---|
| Neuron | -70 mV |
| Skeletal muscle | -90 mV |
| Heart : Ventricles | -90 mV |

**Unstable/oscillating :**

| Pacemakers | Resting membrane potential |
|---|---|
| Heart : SA node | -60 mV to -40 mV |
| GIT : Cajal cells | |
| Respiratory : Pre Botzinger complex (medulla) | |

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**CONTRIBUTING FACTORS :**

1. Diffusion of $K^+$ ion : – 86 mV.
2. $Na^+$–$K^+$ pump : –4 mV.
3. Diffusion of $Na^+$ ion : Very less.

**CLINICAL SIGNIFICANCE :**

**1. Hyperkalemia :**

Eg : In rhabdomyolysis (Breakdown of muscle cell).

↑ Extracellular $K^+$

↓

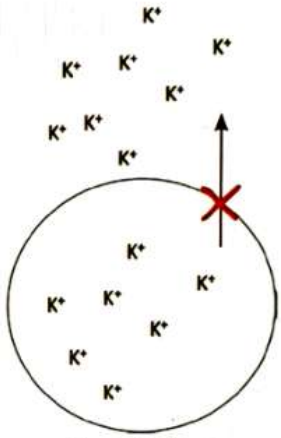
Inhibit passive diffusion of $K^+$

↓

↑ Intracellular $K^+$ (↑ Positive charge)

↓

Depolarized cell (more excitable)




Hyperkalemia

**2. Hypokalemia :**

↓ Extracellular $K^+$

↓

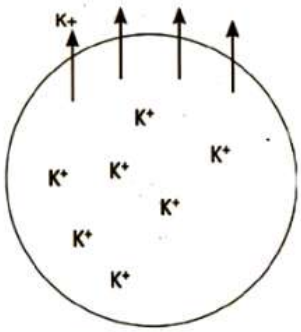
↑ passive diffusion of $K^+$

↓

↓ Intracellular $K^+$ (↑ Negative charge)

↓

Hyperpolarized cell (Less excitable)



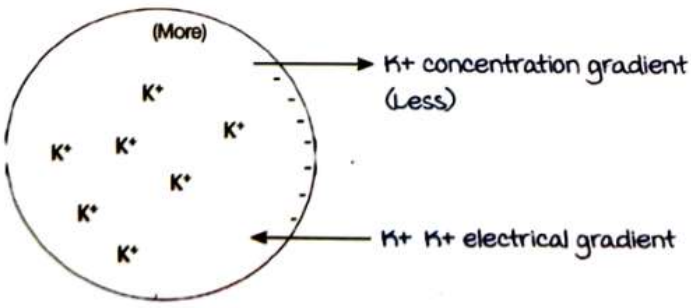

Hypokalemia

## Equilibrium Potential (E)

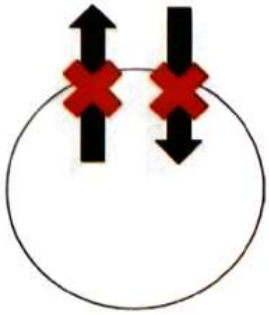
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AKA isoelectric potential (No ion movement).

At EP : Concentration gradient = Electrical gradient.




Isoelectric potential



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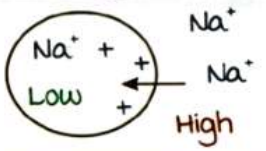
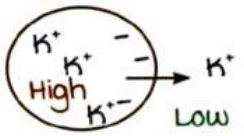
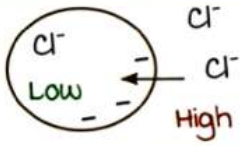
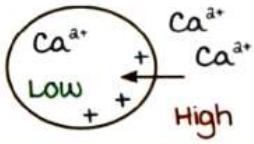


## DETERMINING FACTORS :

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- movement of ions from high concentration to low concentration.
- Charge inside cell at equilibrium.

## VALUES

| Ions | Potential | Diagram |
|---|---|---|
| $Na^+$ | +60 mV |   |
| $K^+$ | −90 mV |   |
| $Cl^-$ Same as RMP of neuron | −70 mV |   |
| $Ca^{a+}$ | +130 mV (Highest) |   |

## EQUATIONS DETERMINING EP :

### Nernst equation :

To calculate equilibrium potential of single ion.

$$E = \pm 61 \log \frac{Co}{Ci} \quad (OR) \quad E = \pm 61 \log \frac{Ci}{Co}$$

Limitation : Single ion considered.

### Goldman's equation :

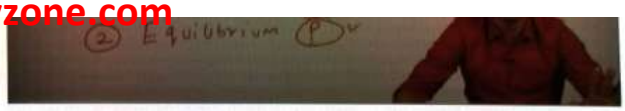
- Consider multiple ions based on :
  a. Concentration.
  b. Permeability.

Applied aspect : Rapid changes in permeability of multiple ions → Generation of action potential.

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## Driving Force (DF)

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**EQUATION :**

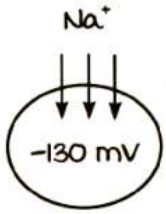
DF = RMP − EP.

Eg: DF ($Na^+$) = − 70 − (+60 mV) = − 130 mV.

**IMPLICATION :**

Driving force : Negative (Intracellular)

- Cation ($Na^+$) → Enter cells
- Anion ($Cl^-$) → Moves out of cells

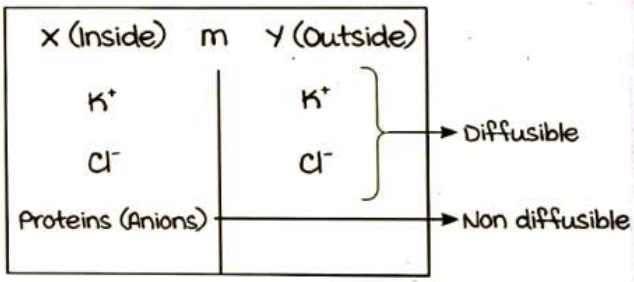



**GIBBS DONNAN EFFECT :**

Presence of non-diffusible ions (Proteins) affects the distribution of diffusible ions.

**Implication :**

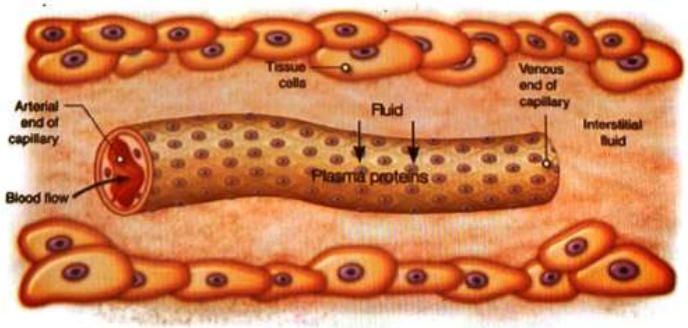
In neuron : RMP is negative d/t non-diffusible protein.




In blood vessel :

Physiology : Colloid oncotic pressure (πc) d/t plasma proteins → Reabsorb fluids.

Pathology : Loss of πc (Kidney diseases) → Interstitial fluid → Edema.




Blood vessel

Feedback